

Module 3 SQL Query Analysis

Modern Software Concepts in Python

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This report presents the results, raw SQL queries, and brief explanations for the nine required analyses and two original database questions.

Question 1

Question

How many entries are from applicants who applied for Fall 2026?

Final Result

Fall 2026 applicant count: 29,576

SQL Query

```
SELECT COUNT(*)  
FROM applicants  
WHERE LOWER(TRIM(term)) = 'fall 2026';
```

Explanation

The query standardizes the term with LOWER and TRIM, filters the table to Fall 2026, and counts every matching row.

Question 2

Question

Among entries that provide a nationality classification, what percentage are international students?

Final Result

Percent international: 46.53%

SQL Query

```
SELECT
100.0
* COUNT(*) FILTER (
WHERE LOWER(TRIM(us_or_international)) = 'international'
)
/ NULLIF(
COUNT(*) FILTER (
WHERE us_or_international IS NOT NULL
AND TRIM(us_or_international) <> ''
),
0
)
FROM applicants;
```

Explanation

The numerator counts entries classified as International. The denominator includes every nonblank nationality classification, including American and Other, while excluding missing values. NULLIF prevents division by zero.

Question 3

Question

What are the average GPA, GRE Quantitative, GRE Verbal, and GRE Analytical Writing scores of applicants who provide each metric?

Final Result

Average GPA: 3.77
Average GRE Quantitative: 165.83
Average GRE Verbal: 160.70
Average GRE Analytical Writing: 4.35

SQL Query

```
SELECT  
AVG(gpa),  
AVG(gre),  
AVG(gre_v),  
AVG(gre_aw)  
FROM applicants;
```

Explanation

Each AVG calculation operates independently. PostgreSQL ignores NULL values, so an applicant contributes to any average for which that applicant supplied a usable value.

Question 4

Question

What is the average GPA of American applicants who applied for Fall 2026?

Final Result

Average GPA of American Fall 2026 applicants: 3.79

SQL Query

```
SELECT AVG(gpa)
FROM applicants
WHERE LOWER(TRIM(term)) = 'fall 2026'
AND LOWER(TRIM(us_or_international)) = 'american'
AND gpa IS NOT NULL;
```

Explanation

The query simultaneously restricts records to Fall 2026, American applicants, and records with a usable GPA. AVG then calculates the mean of those matching GPA values.

Question 5

Question

What percentage of Fall 2025 entries are acceptances?

Final Result

Fall 2025 acceptance percentage: 47.92%

SQL Query

```
SELECT
100.0
* COUNT(*) FILTER (
WHERE LOWER(TRIM(status)) LIKE 'accepted%'
)
/ NULLIF(COUNT(*), 0)
FROM applicants
WHERE LOWER(TRIM(term)) = 'fall 2025';
```

Explanation

The main WHERE clause defines all Fall 2025 entries as the denominator. The filtered count includes statuses beginning with Accepted as the numerator.

Question 6

Question

What is the average GPA of accepted applicants who applied for Fall 2026?

Final Result

Average GPA of accepted Fall 2026 applicants: 3.76

SQL Query

```
SELECT AVG(gpa)
FROM applicants
WHERE LOWER(TRIM(term)) = 'fall 2026'
AND LOWER(TRIM(status)) LIKE 'accepted%'
AND gpa IS NOT NULL;
```

Explanation

The query requires Fall 2026, an acceptance status, and a nonmissing GPA before calculating the average.

Question 7

Question

How many entries are from applicants who applied to Johns Hopkins University for a master's degree in Computer Science?

Final Result

Johns Hopkins Computer Science master's count: 8

SQL Query

```
SELECT COUNT(*)
FROM applicants
WHERE (
  program ILIKE '%Johns Hopkins University%'
  OR program ~* '^[^[:alnum:]]JHU([[:alnum:]]|$)'
)
AND program ILIKE '%Computer Science%'
AND LOWER(TRIM(degree)) = 'masters';
```

Explanation

The query uses the original program and degree fields. It recognizes both Johns Hopkins University and the standalone abbreviation JHU, requires Computer Science, and restricts the degree to Masters.

Question 8

Question

How many Fall 2026 entries are acceptances from applicants applying for a Computer Science PhD at Georgetown, MIT, Stanford, or Carnegie Mellon using the original fields?

Final Result

Original-field count: 28

SQL Query

```
SELECT COUNT(*)
FROM applicants
WHERE LOWER(TRIM(term)) = 'fall 2026'
AND LOWER(TRIM(status)) LIKE 'accepted%'
AND LOWER(TRIM(degree)) = 'phd'
AND program ILIKE '%Computer Science%'
AND (
  program ILIKE '%Georgetown University%'
OR program ILIKE '%Massachusetts Institute of Technology%'
OR program ~* '^(^[:alnum:]]MIT(^[:alnum:]]|$)'
OR program ILIKE '%Stanford University%'
OR program ILIKE '%Carnegie Mellon University%'
);
```

Explanation

The query requires all five conditions at the same time: Fall 2026, acceptance, PhD, Computer Science, and one of the four listed universities. University and program matching use the original combined program field.

Question 9

Question

Repeat Question 8 using the LLM-generated program and university fields, and compare the results.

Final Result

Original-field count: 28

LLM-field count: 28

Difference: +0

SQL Query

```
SELECT COUNT(*)
FROM applicants
WHERE LOWER(TRIM(term)) = 'fall 2026'
AND LOWER(TRIM(status)) LIKE 'accepted%'
AND LOWER(TRIM(degree)) = 'phd'
AND llm_generated_program ILIKE '%Computer Science%'
AND (
  llm_generated_university ILIKE '%Georgetown University%'
OR llm_generated_university ILIKE '%George Town University%'
OR llm_generated_university
  ILIKE '%Massachusetts Institute of Technology%'
OR llm_generated_university
  ~* '^(^[:alnum:]]MIT([^[:alnum:]]|$)'
OR llm_generated_university ILIKE '%Stanford University%'
OR llm_generated_university ILIKE '%Carnegie Mellon University%'
);
```

Explanation

This query keeps the original term, status, and degree fields but identifies the program and university through the LLM-generated fields. The counts are equal, suggesting that the LLM normalization did not change the qualifying set for this specific analysis. Individual labels can still contain normalization errors even when the final count is unchanged.

Question 10

Question

How do Fall 2026 acceptance percentages compare between American and international applicants?

Final Result

American: 5,955 accepted out of 15,443 entries (38.56%)

International: 4,666 accepted out of 13,492 entries (34.58%)

SQL Query

```
SELECT
us_or_international AS applicant_group,
COUNT(*) AS total_entries,
COUNT(*) FILTER (
WHERE LOWER(TRIM(status)) LIKE 'accepted%'
) AS accepted_entries,
ROUND(
100.0
* COUNT(*) FILTER (
WHERE LOWER(TRIM(status)) LIKE 'accepted%'
)
/ NULLIF(COUNT(*), 0),
2
) AS acceptance_percentage
FROM applicants
WHERE LOWER(TRIM(term)) = 'fall 2026'
AND LOWER(TRIM(us_or_international))
IN ('american', 'international')
GROUP BY us_or_international
ORDER BY applicant_group;
```

Explanation

The query groups Fall 2026 records by nationality classification. For each group, it counts all entries and accepted entries, then calculates the acceptance percentage. These self-reported percentages describe the GradCafe data and should not be treated as official population acceptance rates.

Question 11

Question

Which five universities have the most reported Fall 2026 acceptances?

Final Result

Columbia University: 210
University of Toronto: 210
University of Oxford: 187
University of California, Berkeley: 175
University of Washington: 154

SQL Query

```
SELECT
  llm_generated_university AS university,
  COUNT(*) AS accepted_entries
FROM applicants
WHERE LOWER(TRIM(term)) = 'fall 2026'
AND LOWER(TRIM(status)) LIKE 'accepted%'
AND llm_generated_university IS NOT NULL
AND TRIM(llm_generated_university) <> ''
GROUP BY llm_generated_university
ORDER BY accepted_entries DESC, university
LIMIT 5;
```

Explanation

The query groups accepted Fall 2026 records by the normalized university field, orders the groups by their reported acceptance counts, and returns the five largest groups. The results measure reporting frequency in this dataset rather than official university acceptance totals.